

Section 1: Wavelet Transformation

1. What does CWT stand for?
 - A. Continuous Wavelet Transform
 - B. Discrete Wavelet Transform
 - C. Wavelet Packet Transform
 - D. Inverse Wavelet TransformCorrect Answer: A. Continuous Wavelet Transform
2. Wavelet transformation provides analysis in which two domains?
 - A. Time and frequency
 - B. Space and color
 - C. Magnitude and phase
 - D. Gradient and LaplacianCorrect Answer: A. Time and frequency
3. Which type of signals is wavelet transformation especially useful for?
 - A. Stationary signals
 - B. Non-stationary signals
 - C. Periodic signals only
 - D. Binary signals onlyCorrect Answer: B. Non-stationary signals
4. In CWT, the wavelet function is:
 - A. Only translated
 - B. Only dilated
 - C. Dilated and translated
 - D. Neither dilated nor translatedCorrect Answer: C. Dilated and translated
5. Which wavelet is mentioned for analyzing oscillatory components?
 - A. Haar
 - B. Daubechies
 - C. Morlet
 - D. SymletCorrect Answer: C. Morlet

6. Which wavelet families are used in DWT?
- A. Morlet, Mexican Hat, Shannon
 - B. Daubechies, Haar, Symlet
 - C. Gabor, Fourier, Laplace
 - D. Sobel, Canny, LoG
- Correct Answer: B. Daubechies, Haar, Symlet

7. WPT is an extension of:
- A. CWT
 - B. DWT
 - C. Inverse Wavelet Transform
 - D. Fourier Transform
- Correct Answer: B. DWT

8. The Inverse Wavelet Transform is crucial for:
- A. Edge detection
 - B. Image reconstruction
 - C. Hough voting
 - D. SIFT matching
- Correct Answer: B. Image reconstruction

9. Which application is NOT mentioned for DWT?
- A. Image compression
 - B. Image denoising
 - C. Feature extraction
 - D. Line detection
- Correct Answer: D. Line detection

10. WPT is useful for analyzing signals with:
- A. Simple frequency content
 - B. Complex frequency content
 - C. No frequency content
 - D. Only binary content
- Correct Answer: B. Complex frequency content
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Section 2: Boundary Detection

11. Boundary detection is also known as:

- A. Edge detection
- B. Corner detection
- C. Blob detection
- D. Circle detection

Correct Answer: A. Edge detection

12. Edges represent significant changes in:

- A. Intensity or color
- B. Only color
- C. Only intensity
- D. Only texture

Correct Answer: A. Intensity or color

13. Which task is NOT mentioned as using edges?

- A. Object recognition
- B. Image segmentation
- C. Feature extraction
- D. Audio compression

Correct Answer: D. Audio compression

14. Sobel edge detection is based on:

- A. Gradient
- B. Laplacian
- C. Zero-crossings
- D. Hough voting

Correct Answer: A. Gradient

15. How many convolution kernels does Sobel use?

- A. One
- B. Two
- C. Three
- D. Four

Correct Answer: B. Two

16. What is the size of Sobel kernels?

- A. 2×2

B. 3×3

C. 5×5

D. 7×7

Correct Answer: B. 3×3

17. Which Sobel kernel detects vertical edges?

A. G_x

B. G_y

C. G_z

D. G

Correct Answer: A. G_x

18. Which Sobel kernel detects horizontal edges?

A. G_x

B. G_y

C. G_z

D. G

Correct Answer: B. G_y

19. The Sobel gradient magnitude is calculated as:

A. $G_x + G_y$

B. $G_x^2 + G_y^2$

C. $\sqrt{G_x^2 + G_y^2}$

D. $|G_x| + |G_y|$

Correct Answer: C. $\sqrt{G_x^2 + G_y^2}$

20. Why is CV_64F used in Sobel?

A. To preserve negative values

B. To increase brightness

C. To reduce noise

D. To detect circles

Correct Answer: A. To preserve negative values

21. Why are grayscale images often used in computer vision tasks?

A. To simplify intensity analysis

B. To increase color information

C. To detect circles

D. To perform Hough voting

Correct Answer: A. To simplify intensity analysis

22. What is the correct order of Canny edge detection steps?

- A. Gradient → Gaussian smoothing → NMS → Hysteresis
- B. Gaussian smoothing → Gradient → NMS → Hysteresis
- C. NMS → Gradient → Gaussian smoothing → Hysteresis
- D. Hysteresis → NMS → Gradient → Gaussian smoothing

Correct Answer: B. Gaussian smoothing → Gradient → NMS → Hysteresis

23. What is the purpose of Gaussian smoothing in Canny?

- A. Reduces noise to prevent false edges
- B. Increases edge strength
- C. Thins edges
- D. Links weak edges

Correct Answer: A. Reduces noise to prevent false edges

24. What does Non-Maximum Suppression do in Canny?

- A. Reduces noise
 - B. Thins edges to 1-pixel width
 - C. Links strong and weak edges
 - D. Computes gradient magnitude
- Correct Answer: B. Thins edges to 1-pixel width

25. In hysteresis thresholding, weak edges are kept only if:

- A. They are above the high threshold
- B. They are connected to strong edges
- C. They are isolated
- D. They are below the low threshold

Correct Answer: B. They are connected to strong edges

26. Canny edge detection is known for:

- A. Thick and broken edges
 - B. Thin and continuous edges
 - C. Only horizontal edges
 - D. Only vertical edges
- Correct Answer: B. Thin and continuous edges

27. LoG stands for:

- A. Laplacian of Gaussian
- B. Line of Gradient
- C. Logarithm of Gradient

D. Length of Gaussian

Correct Answer: A. Laplacian of Gaussian

28. In LoG, edges are indicated by:

A. Peaks

B. Zero-crossings

C. Valleys

D. High gradients

Correct Answer: B. Zero-crossings

29. Which derivative does Canny/Sobel represent?

A. First derivative

B. Second derivative

C. Third derivative

D. Zeroth derivative

Correct Answer: A. First derivative

30. Which derivative does Laplacian represent?

A. First derivative

B. Second derivative

C. Third derivative

D. Zeroth derivative

Correct Answer: B. Second derivative

Section 3: Hough Transform

31. Hough Transform is used to detect:

A. Lines and circles

B. Only lines

C. Only circles

D. Only ellipses

Correct Answer: A. Lines and circles

32. Hough Transform transforms an image from spatial domain to:

A. Frequency domain

B. Parameter space

C. Color space

D. Scale space

Correct Answer: B. Parameter space

33. A line in Cartesian coordinates is represented as:

A. $y = mx + b$

B. $y = ax^2 + bx + c$

C. $x^2 + y^2 = r^2$

D. $y = mx^2 + b$

Correct Answer: A. $y = mx + b$

34. In Hough line detection, the parameter space axes are:

A. x and y

B. m and b

C. a, b, and r

D. ρ and θ only

Correct Answer: B. m and b

35. In Hough Transform, each edge pixel:

A. Votes for possible shapes

B. Is discarded

C. Becomes a keypoint

D. Is smoothed

Correct Answer: A. Votes for possible shapes

36. Peaks in the Hough parameter space represent:

A. Lines or circles

B. Noise

C. Gradients

D. Keypoints

Correct Answer: A. Lines or circles

37. A circle in Cartesian coordinates is represented as:

A. $y = mx + b$

B. $(x - a)^2 + (y - b)^2 = r^2$

C. $x^2 + y^2 = r$

D. $y = ax^2 + bx + c$

Correct Answer: B. $(x - a)^2 + (y - b)^2 = r^2$

38. The parameter space for circle detection has how many dimensions?

A. 1

B. 2

C. 3

D. 4

Correct Answer: C. 3

39. In circle Hough Transform, the parameters are:

A. m and b

B. a, b, and r

C. x and y

D. ρ and θ

Correct Answer: B. a, b, and r

40. Who introduced the Hough Transform?

A. Paul Hough

B. John Canny

C. David Lowe

D. Irvin Sobel

Correct Answer: A. Paul Hough

41. The Hough Transform was first introduced in:

A. 1952

B. 1962

C. 1972

D. 1982

Correct Answer: B. 1962

42. Back-transformation in Hough Transform reveals:

A. Position and orientation of shapes

B. Only color

C. Only texture

D. Only noise

Correct Answer: A. Position and orientation of shapes

Section 4: SIFT Feature Extraction

43. SIFT stands for:

A. Scale-Invariant Feature Transform

- B. Scale-Invariant Fourier Transform
- C. Simple Invariant Feature Transform
- D. Spatial Invariant Feature Transform

Correct Answer: A. Scale-Invariant Feature Transform

44. SIFT is used for:

- A. Object recognition
- B. Image stitching
- C. Tracking
- D. All of the above

Correct Answer: D. All of the above

45. DoG stands for:

- A. Difference of Gaussians
- B. Derivative of Gradients
- C. Difference of Gradients
- D. Derivative of Gaussians

Correct Answer: A. Difference of Gaussians

46. The DoG is calculated as:

- A. $L(x, y, k\sigma) - L(x, y, \sigma)$
- B. $L(x, y, \sigma) + L(x, y, k\sigma)$
- C. $L(x, y, \sigma) \times L(x, y, k\sigma)$
- D. $L(x, y, \sigma) / L(x, y, k\sigma)$

Correct Answer: A. $L(x, y, k\sigma) - L(x, y, \sigma)$

47. Scale-space extrema detection finds:

- A. Local extrema in scale space
- B. Global maxima only
- C. Zero-crossings
- D. Edges only

Correct Answer: A. Local extrema in scale space

48. In keypoint localization, a candidate keypoint is compared with how many neighbors?

- A. 8
- B. 16
- C. 26
- D. 32

Correct Answer: C. 26

49. The 26 neighbors consist of:

- A. 8 same layer, 9 above, 9 below
- B. 9 same layer, 8 above, 9 below
- C. 8 same layer, 8 above, 10 below
- D. 10 same layer, 8 above, 8 below

Correct Answer: A. 8 same layer, 9 above, 9 below

50. Orientation assignment in SIFT makes keypoints invariant to:

- A. Rotation
- B. Scale
- C. Illumination
- D. All of the above

Correct Answer: A. Rotation

51. The descriptor in SIFT is generated from:

- A. 4×4 sub-regions and 8 orientation bins
- B. 3×3 sub-regions and 8 orientation bins
- C. 4×4 sub-regions and 6 orientation bins
- D. 8×8 sub-regions and 8 orientation bins

Correct Answer: A. 4×4 sub-regions and 8 orientation bins

52. The SIFT descriptor size is:

- A. 64
- B. 128
- C. 256
- D. 512

Correct Answer: B. 128

53. SIFT keypoint matching commonly uses:

- A. Euclidean distance
- B. Manhattan distance
- C. Cosine similarity
- D. Hamming distance

Correct Answer: A. Euclidean distance

54. Outlier rejection in SIFT often uses:

- A. RANSAC
- B. Sobel
- C. Canny

D. Hough

Correct Answer: A. RANSAC

55. Homography estimation is used for:

A. Image stitching

B. Object recognition

C. Both A and B

D. Neither A nor B

Correct Answer: C. Both A and B

56. SIFT keypoints are invariant to:

A. Scale and rotation

B. Only scale

C. Only rotation

D. Only translation

Correct Answer: A. Scale and rotation

57. In SIFT, the descriptor is robust to:

A. Light changes

B. Only rotation

C. Only scale

D. Only noise

Correct Answer: A. Light changes

58. The SIFT descriptor is formed by concatenating:

A. Histograms of gradient orientations

B. Histograms of colors

C. Histograms of intensities

D. Histograms of edges

Correct Answer: A. Histograms of gradient orientations

59. Keypoint localization refines location by fitting:

A. 3D quadratic function

B. 2D linear function

C. 3D cubic function

D. 2D quadratic function

Correct Answer: A. 3D quadratic function

60. SIFT detects keypoint candidates at:

A. Multiple scales

- B. Single scale
 - C. Only the finest scale
 - D. Only the coarsest scale
- Correct Answer: A. Multiple scales
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Section 5: OpenCV Functions & Parameters

61. Which `cv2.imread` flag loads an image in grayscale?
- A. `cv2.IMREAD_COLOR`
 - B. `cv2.IMREAD_GRAYSCALE`
 - C. `cv2.IMREAD_UNCHANGED`
 - D. `cv2.IMREAD_ANYCOLOR`
- Correct Answer: B. `cv2.IMREAD_GRAYSCALE`
62. What is the integer value of `cv2.IMREAD_GRAYSCALE`?
- A. 1
 - B. 0
 - C. -1
 - D. 2
- Correct Answer: B. 0
63. Which `cv2.cvtColor` code converts BGR to RGB?
- A. `cv2.COLOR_BGR2GRAY`
 - B. `cv2.COLOR_BGR2RGB`
 - C. `cv2.COLOR_RGB2BGR`
 - D. `cv2.COLOR_BGR2XYZ`
- Correct Answer: B. `cv2.COLOR_BGR2RGB`
64. What is the integer value of `cv2.COLOR_BGR2GRAY`?
- A. 4
 - B. 6
 - C. 8
 - D. 10
- Correct Answer: B. 6
65. In `cv2.Sobel`, which depth is used to preserve negative values?
- A. `cv2.CV_8U`

B. cv2.CV_16S

C. cv2.CV_32F

D. cv2.CV_64F

Correct Answer: D. cv2.CV_64F

66. In cv2.Sobel, what does ksize=3 mean?

A. 3×3 kernel

B. 3×1 kernel

C. 1×3 kernel

D. 3×3 image

Correct Answer: A. 3×3 kernel

67. In cv2.GaussianBlur, the kernel size must be:

A. Positive odd numbers

B. Positive even numbers

C. Any positive number

D. Any integer

Correct Answer: A. Positive odd numbers

68. In the lab, what Gaussian blur kernel size is used?

A. (3, 3)

B. (5, 5)

C. (7, 7)

D. (9, 9)

Correct Answer: B. (5, 5)

69. In the lab, what sigmaX value is used for Gaussian blur?

A. 1.0

B. 1.4

C. 2.0

D. 2.5

Correct Answer: B. 1.4

70. In cv2.Canny, what are the lower and upper thresholds used in the lab?

A. 30, 100

B. 50, 150

C. 100, 200

D. 150, 250

Correct Answer: B. 50, 150

71. In cv2.Canny, what does apertureSize control?

- A. Sobel operator aperture size
- B. Gaussian kernel size
- C. Threshold value
- D. Image depth

Correct Answer: A. Sobel operator aperture size

72. In cv2.Canny, if L2gradient=True, which norm is used?

- A. L1 norm
- B. L2 norm
- C. L^∞ norm
- D. L0 norm

Correct Answer: B. L2 norm

73. In cv2.Laplacian, which depth is used in the lab?

- A. cv2.CV_8U
- B. cv2.CV_16S
- C. cv2.CV_32F
- D. cv2.CV_64F

Correct Answer: D. cv2.CV_64F

74. What does cv2.convertScaleAbs do?

- A. Converts to absolute values and scales to 8-bit
- B. Converts to grayscale
- C. Converts to binary
- D. Converts to float

Correct Answer: A. Converts to absolute values and scales to 8-bit

75. In cv2.convertScaleAbs, what does alpha control?

- A. Scale factor (contrast)
- B. Delta (brightness)
- C. Kernel size
- D. Threshold

Correct Answer: A. Scale factor (contrast)

76. In cv2.HoughCircles, which method is implemented?

- A. cv2.HOUGH_STANDARD
- B. cv2.HOUGH_PROBABILISTIC
- C. cv2.HOUGH_GRADIENT

D. cv2.HOUGH_MULTI_SCALE

Correct Answer: C. cv2.HOUGH_GRADIENT

77. In cv2.HoughCircles, what does minDist represent?

A. Minimum distance between centers of detected circles

B. Minimum radius

C. Maximum radius

D. Accumulator threshold

Correct Answer: A. Minimum distance between centers of detected circles

78. In cv2.HoughCircles, what does param2 represent?

A. Accumulator threshold for circle centers

B. Higher threshold for Canny

C. Minimum radius

D. Maximum radius

Correct Answer: A. Accumulator threshold for circle centers

79. In cv2.SIFT_create, what is the default nOctaveLayers?

A. 1

B. 2

C. 3

D. 4

Correct Answer: C. 3

80. In cv2.SIFT_create, what is the default contrastThreshold?

A. 0.01

B. 0.03

C. 0.04

D. 0.05

Correct Answer: C. 0.04

81. In cv2.SIFT_create, what is the default edgeThreshold?

A. 5

B. 10

C. 15

D. 20

Correct Answer: B. 10

82. In cv2.SIFT_create, what is the default sigma?

A. 1.0

B. 1.4

C. 1.6

D. 2.0

Correct Answer: C. 1.6

83. In `cv2.SIFT_create`, what is the default `descriptorType`?

A. `CV_8U`

B. `CV_16S`

C. `CV_32F`

D. `CV_64F`

Correct Answer: C. `CV_32F`

84. Which flag in `cv2.drawKeypoints` draws keypoints with size and orientation?

A. `cv2.DRAW_MATCHES_FLAGS_DEFAULT`

B. `cv2.DRAW_MATCHES_FLAGS_DRAW_RICH_KEYPOINTS`

C. `cv2.DRAW_MATCHES_FLAGS_DRAW_OVER_OUTIMG`

D. `cv2.DRAW_MATCHES_FLAGS_NOT_DRAW_SINGLE_POINTS`

Correct Answer: B. `cv2.DRAW_MATCHES_FLAGS_DRAW_RICH_KEYPOINTS`

85. What is the shape of SIFT descriptors for N keypoints?

A. $(N, 64)$

B. $(N, 128)$

C. $(128, N)$

D. $(64, N)$

Correct Answer: B. $(N, 128)$

86. Which function detects keypoints and computes descriptors in SIFT?

A. `sift.detect()`

B. `sift.compute()`

C. `sift.detectAndCompute()`

D. `sift.keypoints()`

Correct Answer: C. `sift.detectAndCompute()`

87. In `cv2.HoughLines`, what does `rho` represent?

A. Distance resolution in pixels

B. Angle resolution in radians

C. Accumulator threshold

D. Minimum line length

Correct Answer: A. Distance resolution in pixels

88. In cv2.HoughLines, what does theta represent?

- A. Distance resolution
- B. Angle resolution in radians
- C. Threshold
- D. Maximum angle

Correct Answer: B. Angle resolution in radians

89. In cv2.HoughLines, what does threshold represent?

- A. Accumulator threshold
- B. Minimum line length
- C. Maximum line gap
- D. Angle resolution

Correct Answer: A. Accumulator threshold

90. In cv2.Sobel, what does dx=1, dy=0 compute?

- A. Horizontal gradient (Gx)
- B. Vertical gradient (Gy)
- C. Laplacian
- D. Magnitude

Correct Answer: A. Horizontal gradient (Gx)

91. In cv2.Sobel, what does dx=0, dy=1 compute?

- A. Horizontal gradient (Gx)
- B. Vertical gradient (Gy)
- C. Laplacian
- D. Magnitude

Correct Answer: B. Vertical gradient (Gy)

92. Which function is used to compute gradient magnitude in the lab?

- A. `np.sqrt(sobel_x**2 + sobel_y**2)`
- B. `np.add(sobel_x, sobel_y)`
- C. `np.multiply(sobel_x, sobel_y)`
- D. `np.abs(sobel_x + sobel_y)`

Correct Answer: A. `np.sqrt(sobel_x**2 + sobel_y**2)`

93. What does `cv2.cvtColor(image, cv2.COLOR_BGR2RGB)` do?

- A. Converts BGR to RGB for Matplotlib
- B. Converts RGB to BGR
- C. Converts to grayscale

D. Converts to binary

Correct Answer: A. Converts BGR to RGB for Matplotlib

94. In cv2.GaussianBlur, what happens if sigmaY=0?

A. It is set equal to sigmaX

B. It is set to 1

C. It is set to 0

D. It causes an error

Correct Answer: A. It is set equal to sigmaX

95. In cv2.Laplacian, what is the default ksize?

A. 1

B. 3

C. 5

D. 7

Correct Answer: A. 1

96. Which edge detector uses hysteresis thresholding?

A. Sobel

B. Canny

C. LoG

D. Hough

Correct Answer: B. Canny

97. Which edge detector uses zero-crossings?

A. Sobel

B. Canny

C. LoG

D. Hough

Correct Answer: C. LoG

98. Which transform is used for line and circle detection?

A. Fourier Transform

B. Hough Transform

C. Wavelet Transform

D. SIFT

Correct Answer: B. Hough Transform

99. Which feature detector is scale-invariant?

A. Sobel

B. Canny

C. SIFT

D. Hough

Correct Answer: C. SIFT

100. Which algorithm is used for outlier rejection in SIFT matching?

A. RANSAC

B. Sobel

C. Canny

D. LoG

Correct Answer: A. RANSAC